Module 2: Disease Inheritance and Expression

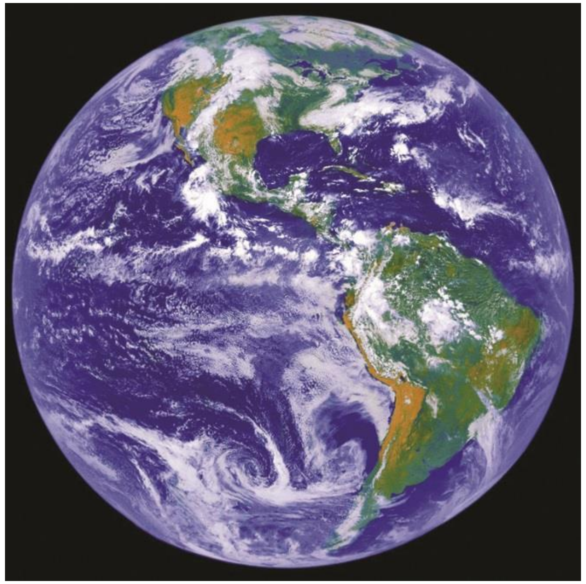
January 19 2026

Outline

- Monogenic Inheritance
- Variable expression of monogenic diseases
- Polygenic Inheritance
- Variable expression of polygenic diseases

What is the definition of 'Genetics'?

- Definition: *the science of heredity, dealing with resemblances and differences of related organisms resulting from the interaction of their genes and the environment* (https://www.dictionary.com/browse/genetics)
- Genetics: The Science of (Nearly) Everything Biological
  - inheritance
  - genes, gene function, gene expression, gene regulation
  - the interactions between genes and environment and how those interactions contribute to disease
  - human biology, evolution, disease
- Why is it relevant?
  - risk for disease
  - prenatal testing, newborn screening, genetic counseling
  - disease treatment and prevention
  - impacts virtually all areas of medicine and biomedical sciences

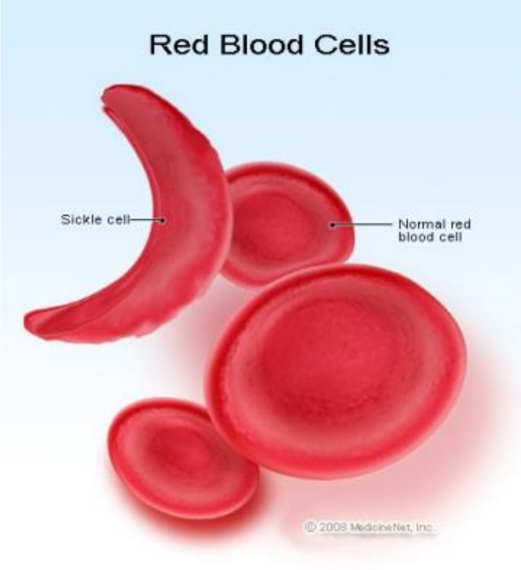


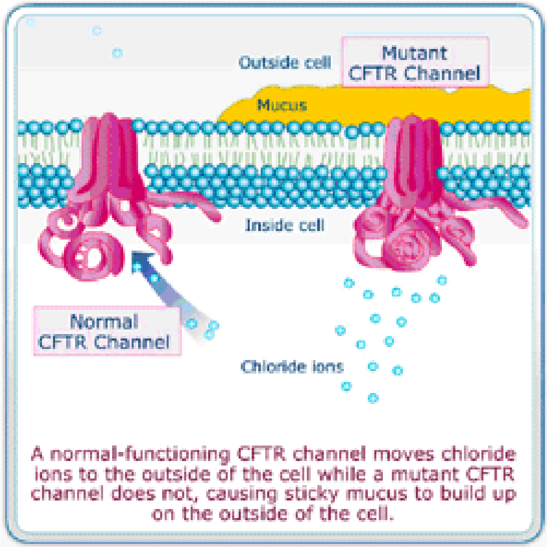
Categories of Heritable Diseases

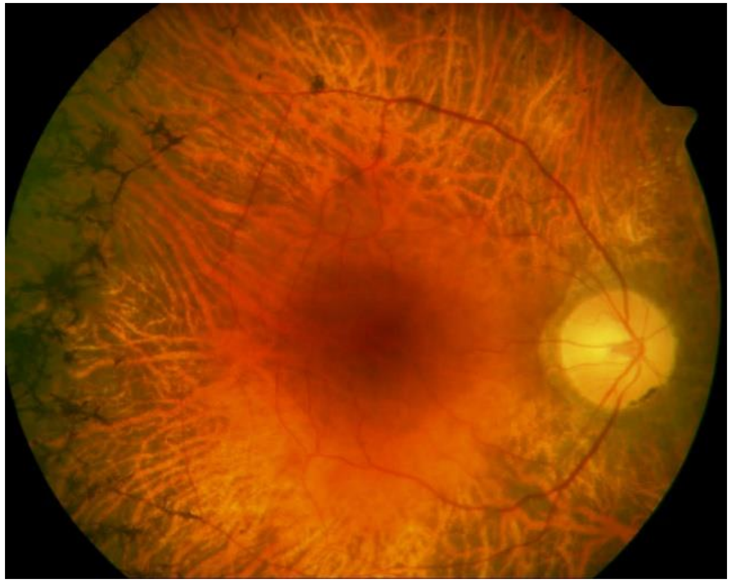
- **Chromosome disorders**: cased by change in dosage of genes located on entire chromosomes or chromosome segments (e.g. Down Syndrome)
- **Monogenic disorder**: caused by pathogenic mutations in individual genes.
- **Polygenic diseases**: results from the combined impact of variants in many different genes

Major Classes of Genetic Disorders

- **Single Gene Disorders**
  - ~10,000 disorders
  - Risk: 1 in 50
  - Mendelian inheritance patterns
  - In an individual a DNA change in a single gene leads to disease
  - One change in one causative gene causes a specific disorder
    - examples: cystic fibrosis, sickle cell disease
  - Locus heterogeneity: same or similar disease caused by change in any one of several genes
    - example: retinitis pigmentosa







Monogenic Inheritance

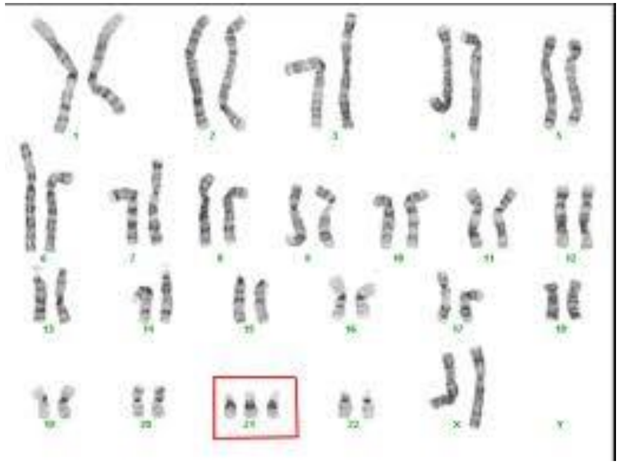
Core Mendelian principles underlying monogenic inheritance patterns:

- **Law of Segregation**: each individual carries two alleles for a given gene, one from each parent, and these alleles segregate during the formation of gametes
    - explains how a single genetic variant (**allele**) from one parent can influence the traits of an offspring
- **Law of Independent Assortment**: the inheritance of one gene does not affect the inheritance of another, as long as they are on separate chromosomes

Major Classes of Genetic Disorders

- **Chromosomal Disorders**
  - Do not have simple Mendelian inheritance patterns
  - Due to altered number of chromosomes or duplication/deletion of large chromosome segments
  - Risk: 1 in 140
  - Half of early spontaneous abortions

  - Example: Trisomy 21 (Down syndrome)
    - extra copy or portion of chromosome 21



Autosomal Dominant Disorders

- Heterozygotes affected
- Homozygotes can be much more severely affected
- Transmission
  - **Affected individuals typically have an affected heterozygous parent**
- ~Equal number of males and females
- Example: Marfan Syndrome
  - Skeletal changes: tall, thin arms and legs with long, thin fingers and toes
  - Cardiovascular issues: aortic dilation/rupture, sudden death
  - Connective tissue disorder
  - Mutations in *fibrillin 1* (*FBN1*)
  - Incidence: 1 in 5000

Isaiah Austin

Aa x aa
Aa x Aa
Aa x AA
AA x aa
AA x AA

Autosomal Dominant Disorder

- Caused by a mutation in a single allele of a gene located on one of the autosomal chromosomes
- An affected person has one normal allele and one mutant allele.
- Both males and females are equally affected because autosomes are not sex-linked.
- Example: Huntington's Disease

Autosomal Recessive Disorders

- Individuals homozygous or biallelic for pathogenic alleles are affected
- Transmission
  - Most affected individuals have unaffected parents
  - All progeny of two affected individuals are affected
- Manifests in males and females equally
- Example: Oculocutaneous Albinisms
  - OCA1-7: probably 7 different autosomal genes
  - loss of pigment in skin, hair and eyes
  - defect in melanin pigment synthesis (tyrosinase)
  - Incidence: 1 in 20,000

Aa x Aa
Aa x aa
aa x aa

Autosomal Recessive Disorder

- Caused by mutations in both alleles of a gene located on an autosomal chromosome (usually leading to a loss of function)
- Individuals with only one mutant allele (**heterozygotes**) are **carriers** and typically asymptomatic.
- **Consanguinity** (mating between close relatives) increases the probability of recessive disorders
- Example: Cystic fibrosis, sickle cell anemia

X-Linked Dominant Disorders

- Anyone with affected allele is typically affected
  - Females heterozygous and (rarely) homozygous
  - Males hemizygous
  - Severity of disease in males > females
  - Incidence of affected females can be > than affected males due to male lethality
- Transmission
  - 50% of children of heterozygous mothers will be affected
  - 100% of daughters of affected fathers will be affected
  - **No male to male transmission**
- Example: X-linked hypophosphatemic rickets (XLHR)
  - low levels of phosphate in blood with poor formation of bones and teeth plus short stature
  - Mutations in *PHEX*
  - Incidence: 1 in 20,000

$XY \times XX^A$

$X^AY \times XX$

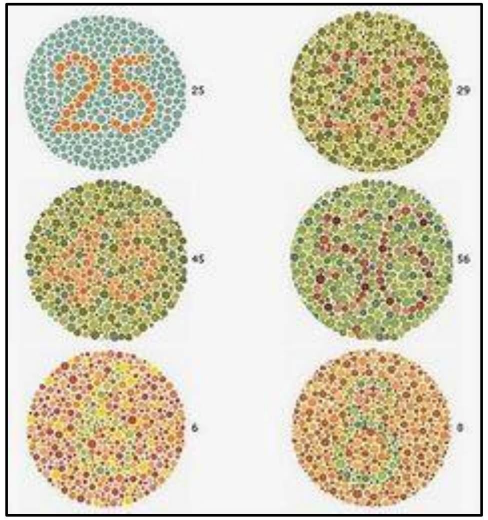
$X^AY \times XX^A$

X-Linked Dominant Disorder

- Caused by a mutation in one allele of a gene located on the X chromosome. One
- copy of the mutant allele in either males or females is enough. Males tend to be
- more severely affected because they have only one X chromosome
- Affected females may have a milder phenotype due to **X-chromosome inactivation**, where some cells randomly inactivate the X chromosome carrying the mutant allele

X-Linked Recessive Disorders

- Homozygous or biallelic females and hemizygous males typically affected
- Carrier females typically unaffected
- Affected males >> affected females
- Transmission
  - No male to male transmission
  - 50% of sons of carrier mothers will be affected
  - 50% of daughters of carrier mothers will be carriers
  - All daughters of affected males are carriers
  - All sons of homozygous affected mothers are affected
  - All daughters of homozygous affected mothers are carriers
- Example: Red-green color vision deficiency (color blindness)
  - Inability to discern red/yellow/green
  - Mutations in *OPN1MW* and *OPN1LW* on X chromosome
  - European incidence: 1/20 males, 1/400 females
  - No obvious effect on fitness



$XY \times XX^a$

$XY \times X^aX^a$

$X^aY \times XX^a$

$X^aY \times X^aX^a$

X-Linked Recessive Disorder

- Primarily expressed in males. Since they have only one X chromosome, a single mutant allele on the X chromosome is enough.
- Females are typically carriers and may not show symptoms due to the presence of a second, normal X chromosome.

Y-Linked Disorders

- Only males with affected Y are affected
- Transmission
  - Theoretically, all sons of affected male will be affected
  - In practice, father-son transmission (i.e. inheritance) is rare
- Most Y-linked traits are related to fertility and/or sexual development
- Individuals with Y-linked mutations are typically sterile
- Newly arising mutations/insertions/deletions in genes on the Y-chromosome are important clinically
- Example: Y-linked retinitis pigmentosa
  - One pedigree described
  - Gene unknown

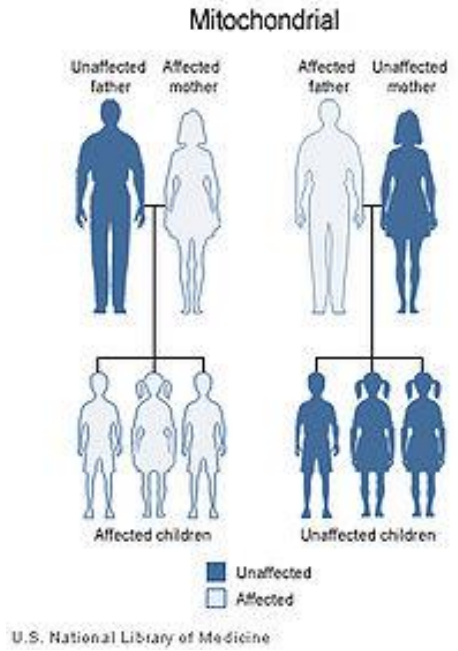
XY^A x XX

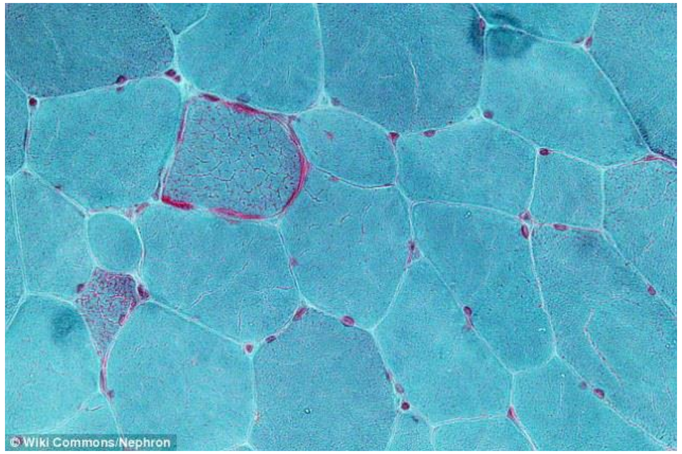
Y-Linked Disorder

- The trait is caused by a mutation in one allele of a gene located on the Y chromosome.
- These traits do not affect females as they do not inherit the Y chromosome.
- **Male-to-male transmission**
- Example: Some forms of male infertility

Major Classes of Genetic Disorders

- **Mitochondrial Inherited Disorders**
  - Special class of single-gene disorders
  - Risk: 1 in 2500
  - Not governed by Mendel's laws
  - Transmitted mother to progeny
  - Never paternally transmitted
  - Caused by mutation in mitochondrial genome
  - Complicated
  - Example: Myoclonic epilepsy with ragged red fibers (MERRF)





Mitochondrial DNA Inheritance

- Mitochondrial inheritance **does not** follow typical Mendelian rules. Disorders
- typically affect high-energy tissues like muscles and the brain due to mitochondria's role in energy production.
- **Matrilineal inheritance**: Affected mothers pass the mtDNA mutation to all of their children (both sons and daughters), but only females pass it on to the next generation. Affected males do not transmit the disorder.
- Examples:
    - **Leber's Hereditary Optic Neuropathy (LHON)**: Causes vision loss due to mitochondrial dysfunction.
    - **Myoclonic Epilepsy with Ragged-Red Fibers (MERRF)**: Affects muscles and the nervous system.

Variable Expression in Genetic Disorders

1. X-chromosomeInactivation
2. 2. Mosaicism
3. 3. Genomic Imprinting
4. 4. Non-Penetrance and Age-Related Penetrance
5. 5. Heteroplasmy

1. X-chromosome Inactivation

- Some cells in females randomly inactivate one of the X chromosomes carrying the mutant allele during early embryonic development
- **Unbalanced/skewed X inactivation** occurs when the fraction of cells in various tissues of carrier females in which the normal or pathogenic allele happens to remain active deviate substantially from the expected 50%

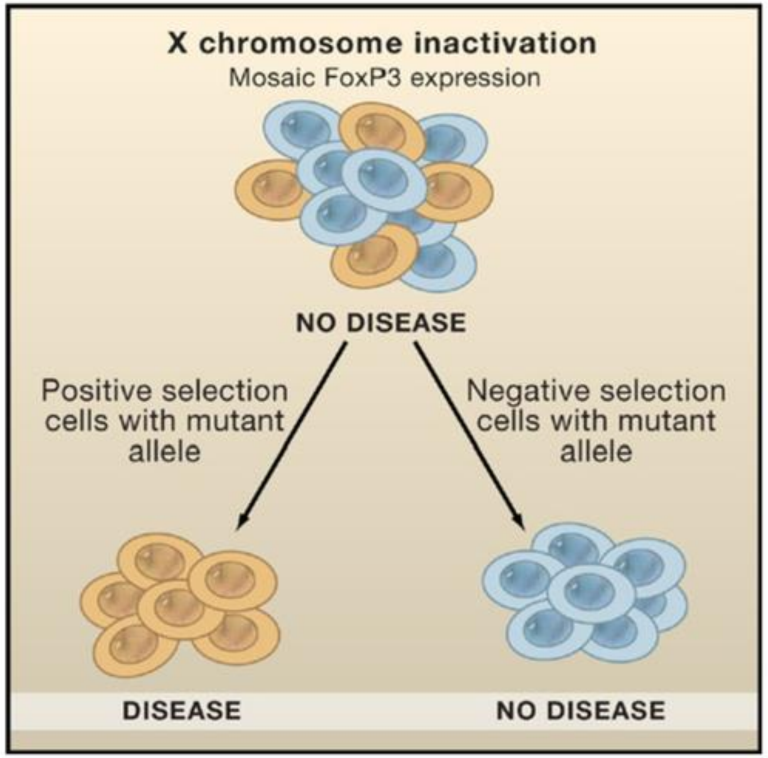
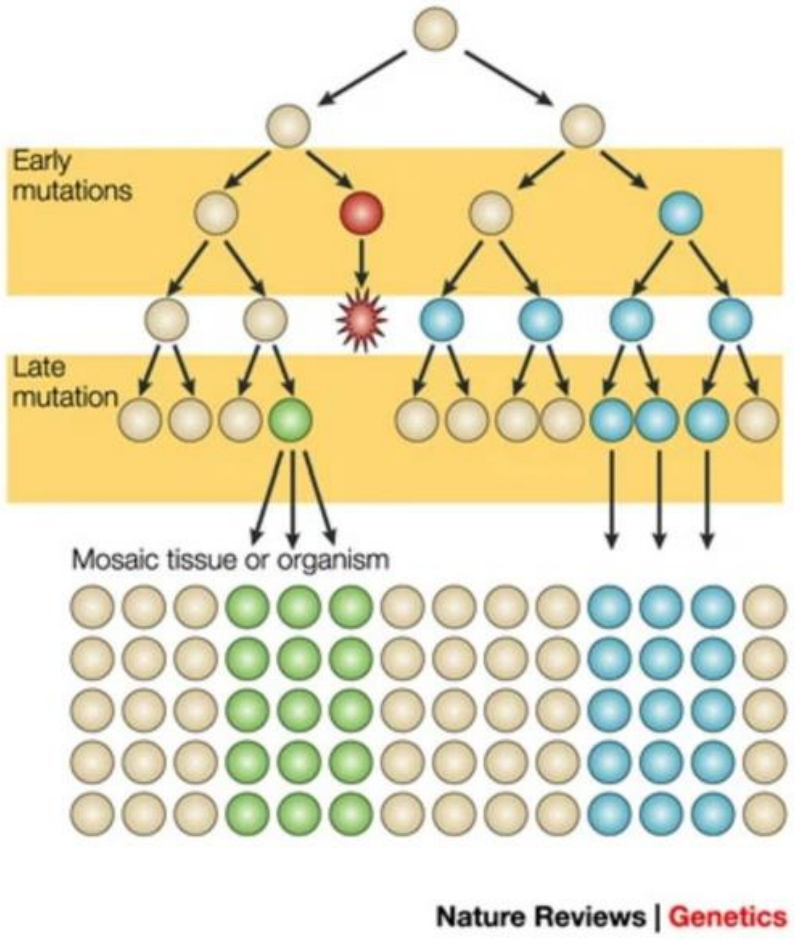


Figure 1. Inactivation of X-Linked Tumor Suppressor Genes and Tumor Formation

Mosaic expression of a mutant X-linked gene, such as FoxP3, in females depends on X chromosome inactivation. Growth of cells carrying a mutant allele can be actively suppressed by negative selection resulting in skewed X inactivation.1

2. Mosaicism

- Occurs when at least two cell lineages that differ genetically but are derived from a single zygote are expressed in an individual or a tissue
- Many types of expression: pure somatic, pure germline, or both
- **Segmental mosaicism**: occurs when the mutation affects morphogenesis and manifests as a segmental or patchy abnormality



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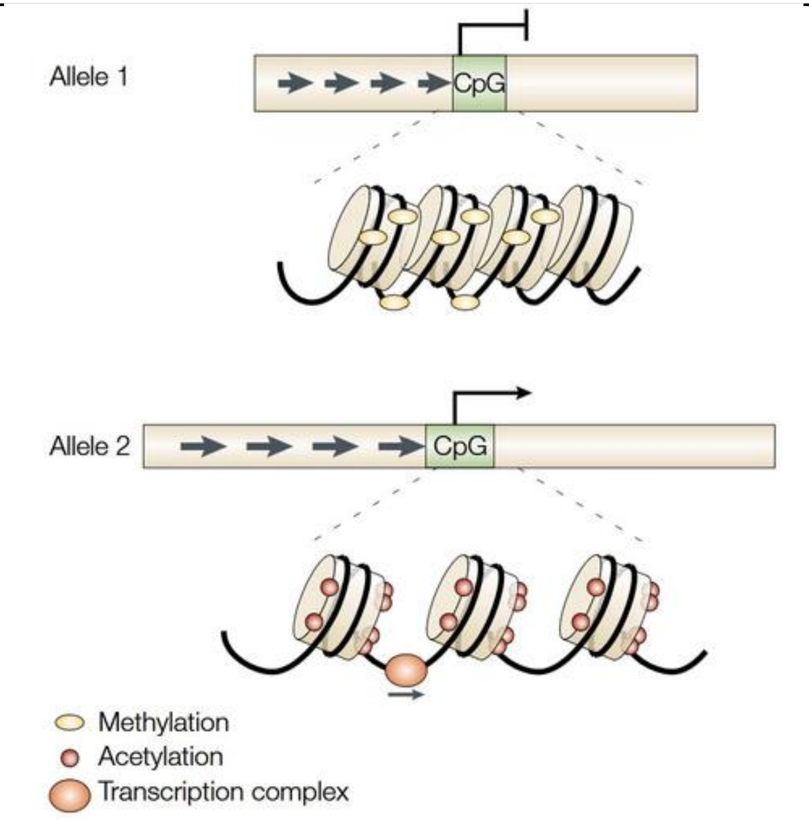
3. Genomic Imprinting

- Occurs when the sex of the parent who transmits the pathogenic allele determines whether there is expression of the disorder in a child.
- Methylation marks the imprinted genes differently in egg and sperm, and inheritance of these epigenetic marks leads to differential gene expression.
- Examples: Prader-Willi syndrome, Angelman syndrome



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4. Non-Penetrance and Age-Related Penetrance

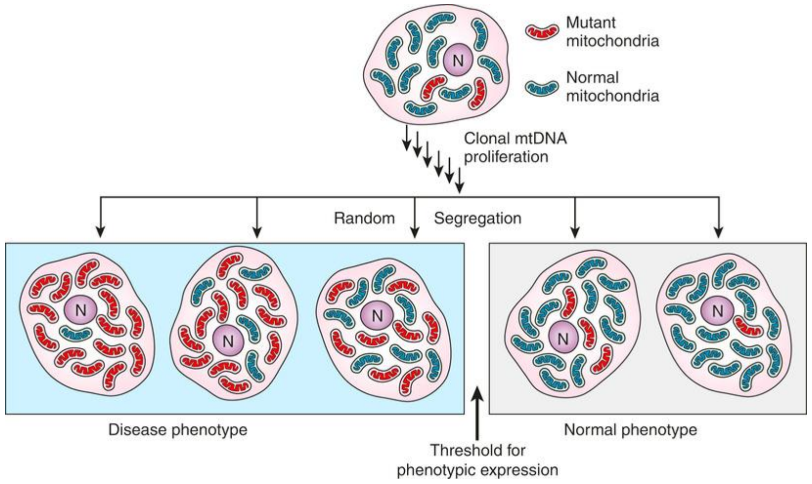
- **Non-Penetrance**: Occurs when an individual carries a disease-causing genetic mutation but does not exhibit any symptoms of the disease.
  - Example: Some carriers of BRCA1/BRCA2 mutations never develop breast or ovarian cancer despite having high-risk mutations.
  - Mechanisms:
    - Other genes that suppress/enhance the expression of the mutation.
    - Lifestyle factors or exposures that mitigate disease development.
    - Epigenetic modifications modulating gene expression without altering DNA sequence.

4. Non-Penetrance and Age-Related Penetrance

- **Age-Related Penetrance**: The likelihood of expressing a genetic trait increases with age.
    - Example: Neurodegenerative symptoms of Huntington's disease typically appear between ages 30 and 50.
    - Mechanisms:
        - Cumulative exposure: Prolonged exposure to environmental factors.
        - Biological aging: Age-related decline in cellular repair mechanisms.
        - Late-onset gene activation: Some genes are expressed or become problematic only later in life.

5. Heteroplasmy

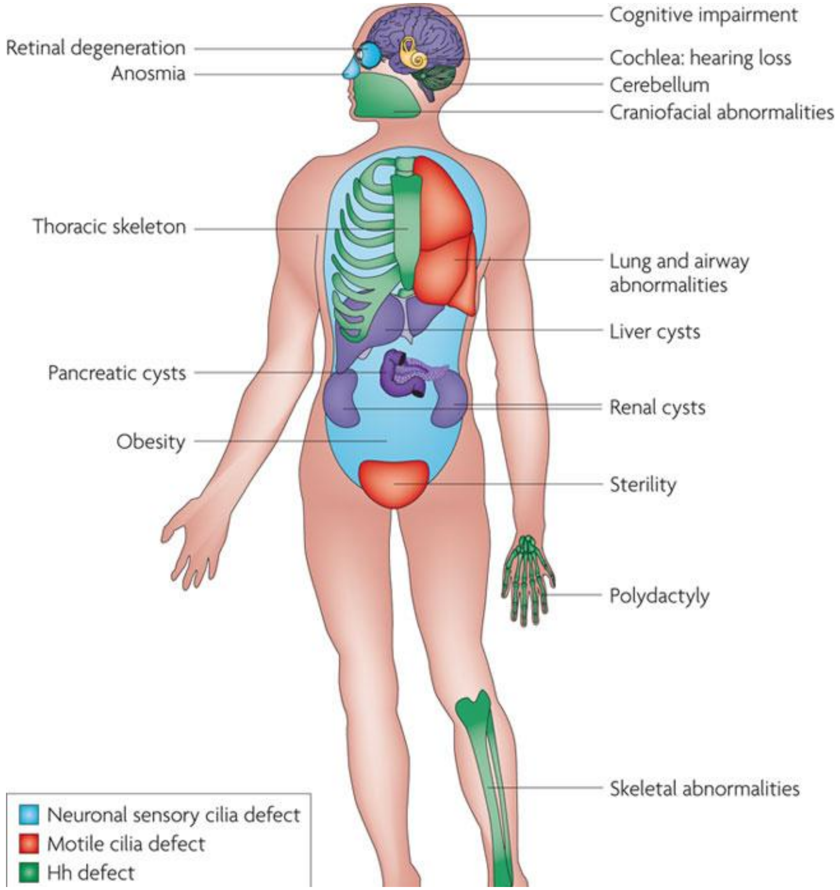
- The expression of a pathogenic variant in mtDNA depends on the relative proportions of normal and variant mtDNA in the cells making up different tissues
- Most pathogenic variants in mtDNA are only present and transmitted in a state of heteroplasmy
    - Exceptions: disorders that are either incompletely penetrant or are not reproductively lethal (e.g. Leber hereditary optic neuropathy)



Major Classes of Genetic Disorders

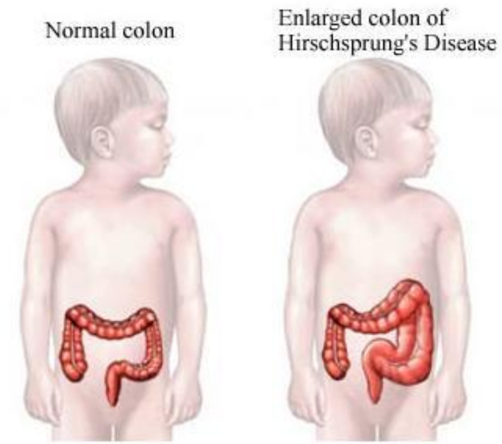
- **Polygenic Disorders**
  - Do not have simple Mendelian inheritance patterns
  - In an individual, changes in more than one gene cause disease
  - Between individuals, changes in different subsets of genes may cause the same disease
  - Each gene may have a small effect
  - Formally do not have a major environmental component

  - Often requires changes in several genes
    - example: some ciliopathies
    - Heart disease and diabetes are referred to as polygenic disorders but strictly speaking they also are influenced by the environment



Major Classes of Genetic Disorders

- **Multifactorial (Complex) Disorders**
  - Do not have simple Mendelian inheritance patterns
  - Likely responsible for majority of human disease
  - Risk: 1 in 20 to 1 in 2
  - Many diseases probably have a multifactorial component
  - Formally defined as diseases due to gene x environment interactions
  - Examples
    - Some forms of Alzheimer disease
    - Hirschsprung disease
    - Heart disease



Polygenic Diseases

- Results from the combined effect of variants in many different genes Each
- variant may cause, protect from, or predispose to a serious defect, often in concert with or triggered by environmental factors
- The same genetic variants can lead to different phenotypic expressions among individuals
    - Does not follow Mendelian inheritance patterns
- Examples: Breast and ovarian cancer, diabetes and inflammatory bowel diseases

Variable Expression through Genetic Factors

- **Allelic Variability**: Different combinations of disease variants can lead to varying disease severity.
- **Gene-Gene Interactions** (Epistasis): One gene's effect can be modified by one or several other genes.
- **Copy Number Variations**: Differences in the number of copies of a particular gene can influence expression.

(We will go through these in more detail in later modules)

Variable Expression through Gene-Environment Interaction

- Environmental factors can modify gene expression and disease risk, so the same combinations of disease variants can lead to different outcomes based on environmental exposures.
- Lifestyle factors: diet, physical activity, substance use
- Physical environment: pollutants, toxins, climate, geography
- Psychosocial factors: stress, socioeconomic status

(We will go through these in more detail in later modules)

Variable Expression through Epigenetic Effects

- **Gene regulation**: Epigenetic modifications can activate or repress genes, leading to variability in disease phenotypes among individuals with similar genetic backgrounds.
- **Environmental interactions**: Lifestyle factors like diet, stress, and exposure to

  toxins
- can induce epigenetic changes influencing disease risk.

  Intergenerational effects: Some epigenetic marks can be transmitted to offspring, affecting their susceptibility to diseases.

(We will go through these in more detail in later modules)

Major Classes of Genetic Disorders

- **Lifetime risk: Multifactorial/Polygenic > Single-Gene/Repeat Disorders > Chromosomal > Mitochondrial**

- Only single-gene disorders follow simple Mendelian inheritance patterns

Individual risks are FYI, but know this

- All other disorders have more complicated patterns of inheritance

- **Usually, all <u>nuclear genes and alleles</u> are transmitted from one generation to the next as predicted by Mendelian principles**

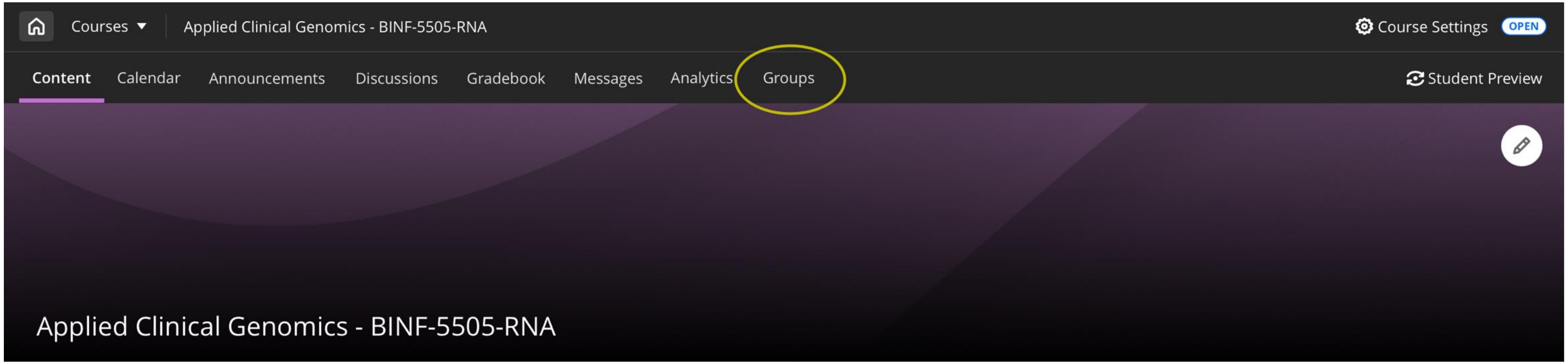
- **Not all <u>inheritance patterns of disorders</u> fit simple Mendelian predictions because**
  - **Disorder is polygenic**
  - **Disorder is multifactorial**
  - **Disorder due to abnormal transmission of chromosomes**
  - **Disorder is mitochondrial**

Mendelian Genetics: Key Terms cont.

- Allele: Alternative forms of a homologous DNA sequence or gene
- Genotype: The allele set at a gene or locus
  - Homozygous: having identical alleles at a locus, e.g. AA or aa
  - Heterozygous: having different alleles at a locus, e.g. Aa
  - Hemizygous: having a single copy of a gene (males - genes on the X & Y)
  - Carrier: heterozygous for recessive pathogenic allele (e.g. Aa)
  - Biallelic: 2 different mutant alleles of the same gene (e.g. a_1a_2)
- Phenotype: Observable feature(s)
  - Physical manifestation of genes

Lab 1a: Navigating Public Databases

The class is assigned into 6 groups of 4-5 students. You can find your group on Blackboard.



- **Lab 1a: Navigating Public Databases**
- Each group is assigned a monogenic/polygenic disease
- For Lab 1a, you will be using the gene associated with your assigned disease. You may
- work with your group but the lab assignment must be submitted individually

[Survey 02]